
DS-306 Data Warehousing and Business Intelligence

Topic 2: Overview of DW and BI

Dr. Khurram Shahzad

Operational Sources (OLTP's)

- Operational computer systems did provide information to run day-to-day operations, and answer's daily questions, but...
 - Also, called **online transactional processing system** (OLTP)
 - Data is read or manipulated with **each** transaction
 - Transactions/queries are simple, and easy to write
 - Usually for middle management
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Typical decision queries

- Decision-making require **complex questions** from **integrated data**
- **Enterprise-wide** data is desired
- Decision makers want to know:
 - ❑ Where to build new oil warehouse?
 - ❑ Which market they should strengthen?
 - ❑ Which customer groups are most profitable?
 - ❑ How much is the total sale by month/ year/ quarter for each offices?
 - ❑ Is there any relation between promotion campaigns and sales growth?
- **Can OLTP answer all such questions, efficiently?**

Failure of old OLTPs

- Inability to **provide** strategic information
- IT receive too many **ad hoc** requests
- Requests are not only numerous, they **change** overtime, because for more understanding more reports
- Users are in **spiral of reports**
- Users have to **depend** on IT for information
- Can't provide enough performance, **slow**
- Strategic information have to be **flexible** and conductive, **analysis driven** (not report driven analysis)

Expectations of new soln.

- DB designed **for** analytical tasks
- Data from **multiple** applications
- **Easy** to use
- Ability of **what-if** analysis, (analysis driven)
- **Read-intensive** data usage
- **Direct** interaction with system, without IT assistance
- **Periodical updating** contents & stable
- Current & **historical** data
- Ability for users to **initiate** reports

DW meets expectations

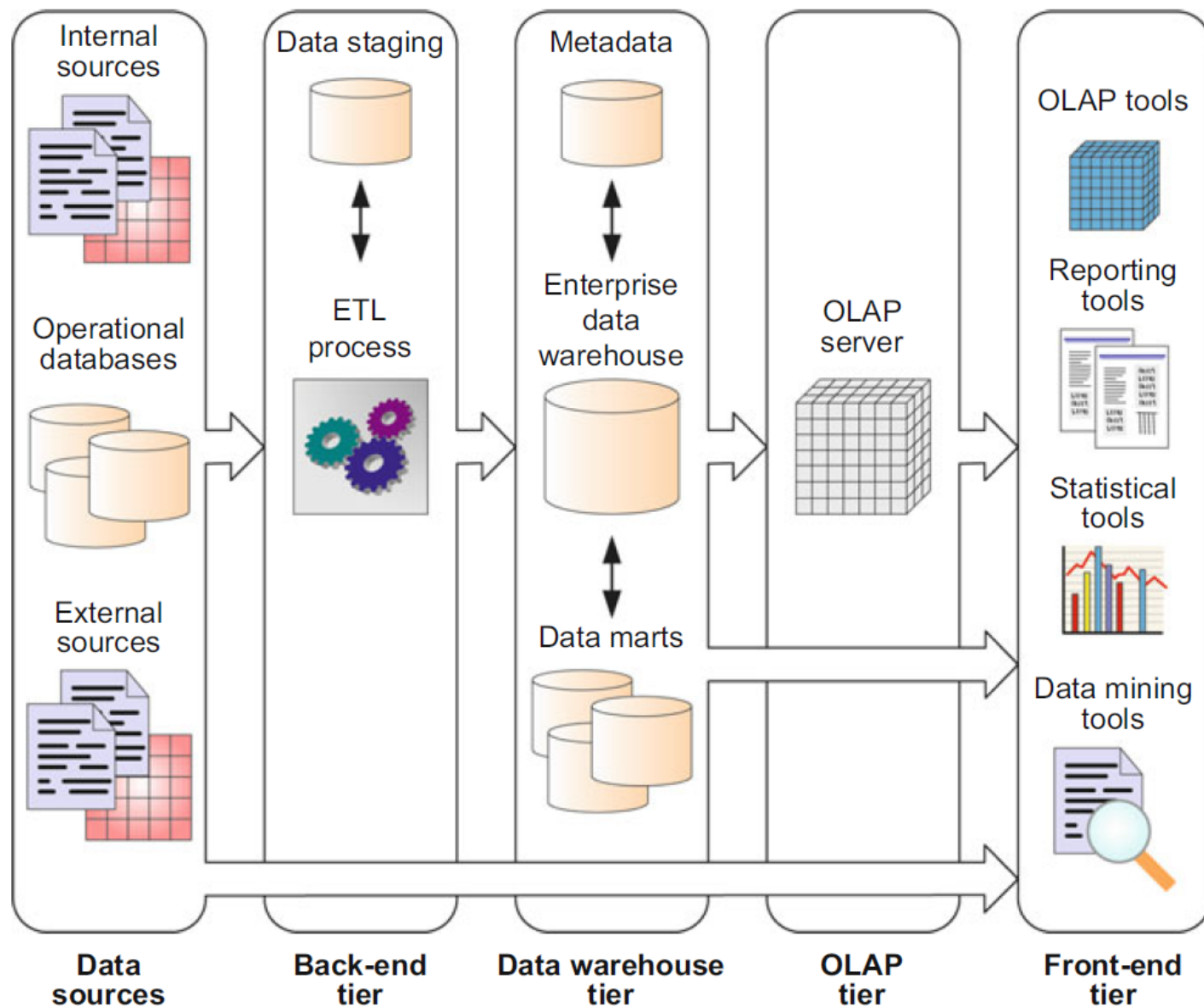
- Provides **enterprise** view
- Current & **historical** data available
- **Decision-transaction** possible without affecting (locking) operational source
- **Reliable** source of information
- Ability for users to **initiate** reports
- Acts as a data source for all **analytical** applications

OLTP vs. BI

Trait	OLTP	BI
User	Middle management	Executives, decision-makers
Function	For day-to-day operations	For analysis & decision support
DB (modeling)	E-R based, after normalization	Star oriented schemas
Data	Current, Isolated	Archived, derived, summarized
Unit of work	Transactions	Complex query
Access, type	DML, read	Read
Access frequency	Very high	Medium to Low
Records accessed	Tens to Hundreds	Thousands to Millions
Quantity of users	Thousands	Very small amount
Usage	Predictable, repetitive	Ad hoc, random, heuristic
DB size	100 MB-GB	100GB-TB
Response time	Sub-seconds	Up-to min.

Architecture of BI solution

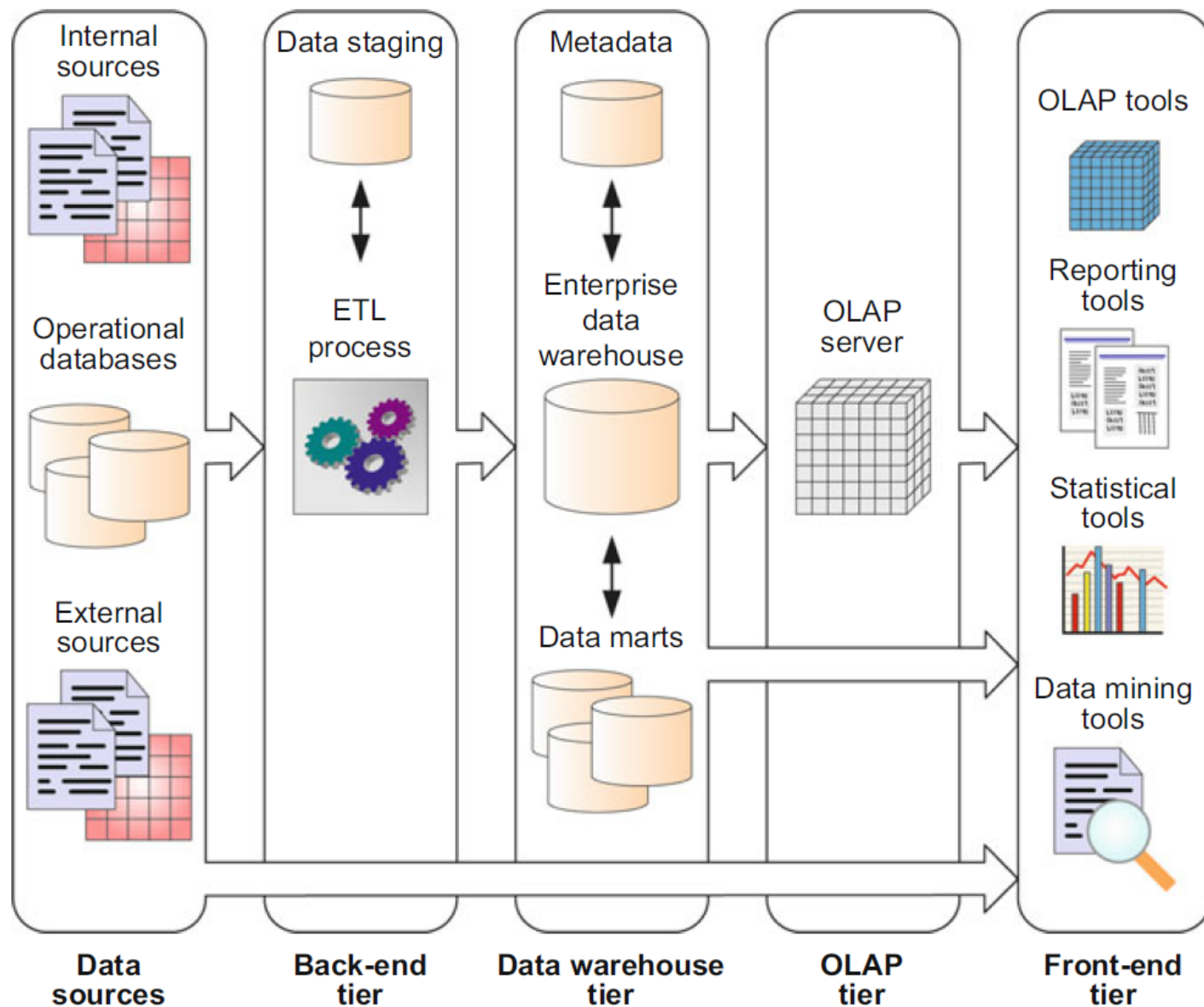
Typical DW architecture



1. Data Sources

- Source data can be grouped into 4 components
 - Production data
 - Comes from operational systems of enterprise
 - Some segments are selected from it
 - Narrow scope, e.g. order details
 - Internal data
 - Private datasheet, documents, customer profiles etc.
 - E.g. Customer profiles for specific offering
 - Special strategies to transform 'IT' to DW (text document)
 - Archived data
 - Old data is archived
 - DW have snapshots of historical data
 - External data
 - Executives depend upon external sources
 - E.g. market data of competitors, car rental require new manufacturing. Define conversion

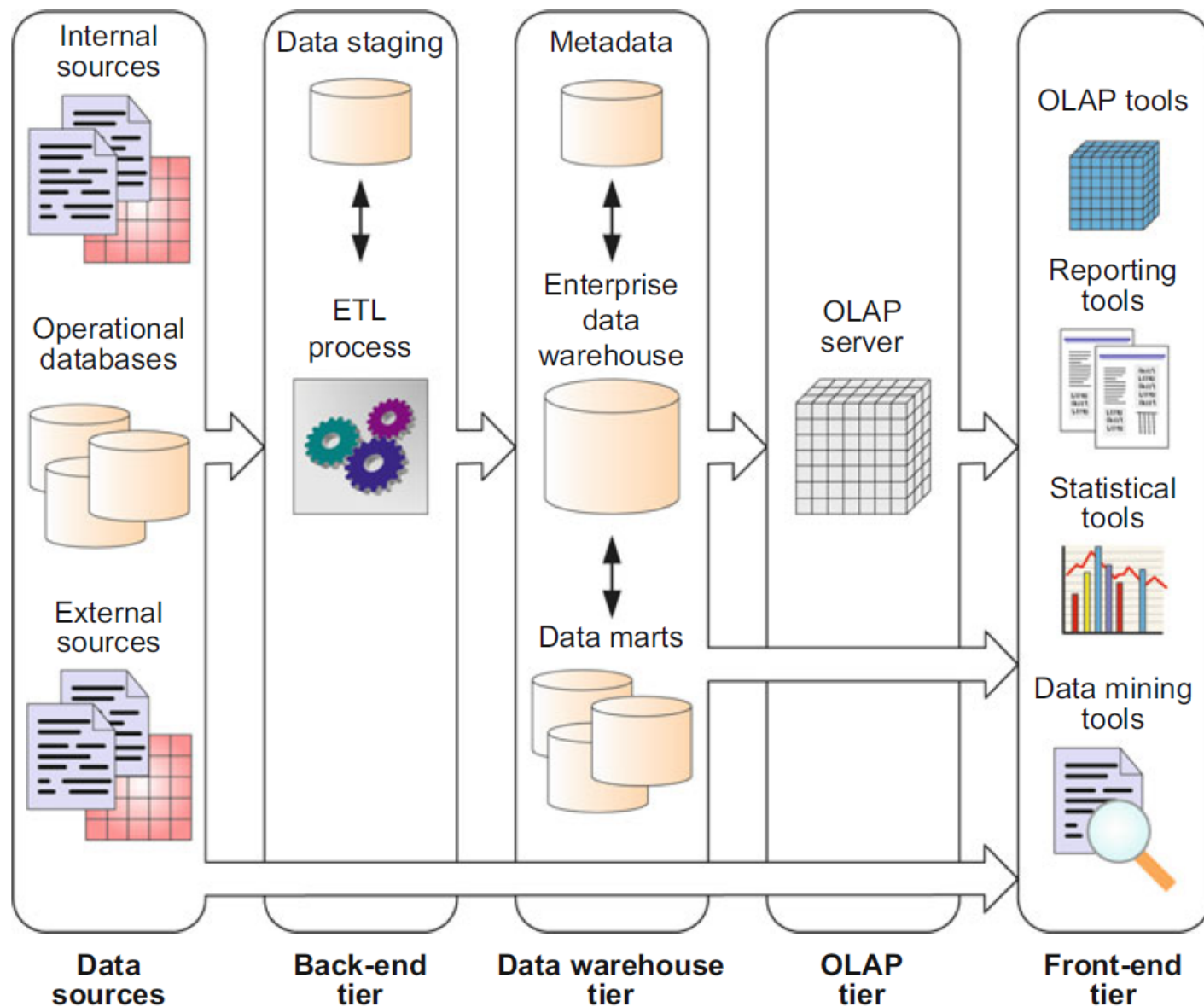
Typical DW architecture



2. Back-end Tier

- After data is extracted, data is to be prepared
- Data extracted from sources needs to be changed, converted and made ready in suitable format
- Three major functions to make data ready
 - Extract
 - Transform
 - Load
- Staging area provides a place and area with a set of functions to
 - Clean
 - Change
 - Combine
 - Convert

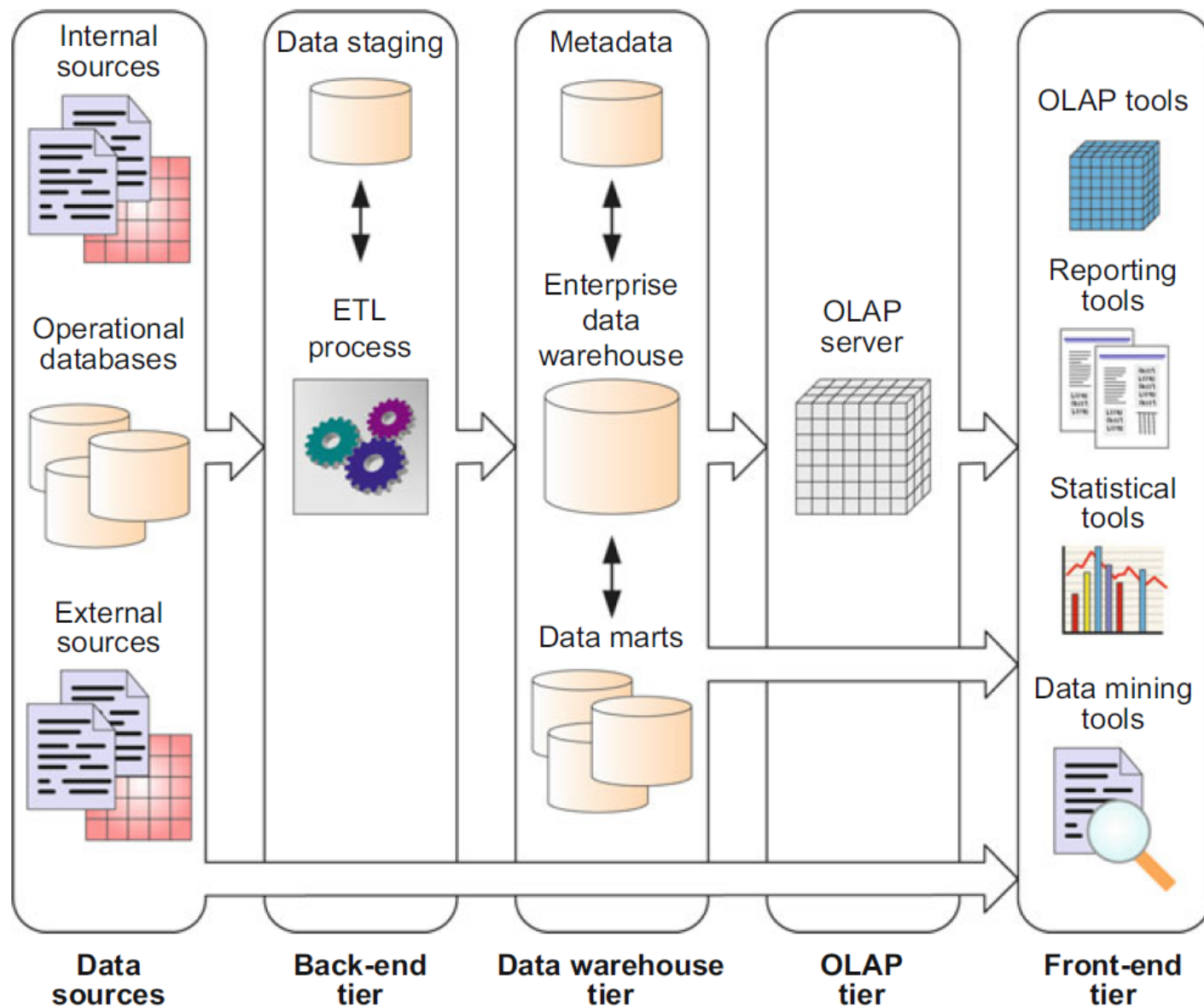
Typical DW architecture



3. Data Warehouse Tier

- Separate repository
- Data structured for efficient processing
- Redundancy is increased
- Updated after specific periods
- Only read-only

Typical DW architecture



4. Information Delivery Component

- Authentication issues
- Active monitoring services
 - Performance
 - User performance
 - Aggregate awareness
 - E.g. mining, OLAP etc

Definition of DW

Data Warehouse

- A data warehouse is a particular database targeted toward decision support.
- It takes data from **various operational databases** and other data sources and **transforms** it into **new structures** that **fit** better for the task of performing business analysis.
- DWs are based on a **multidimensional model**, where data are represented as **dimensions** corresponding to the various business perspectives and cube cells containing the **measures** to be analyzed.

Definition of DW

Inmon defined

“A DW is a subject-oriented, integrated, non-volatile, time-variant collection of data in favor of decision-making”.

Kelly said

“Separate available, integrated, time-stamped, subject-oriented, non-volatile, accessible”

Four properties of DW

Subject-oriented

- In operational sources data is organized by applications, or business processes.
- In DW, subject is the organization method
- Subjects vary with enterprise
- These are critical factors, that affect performance
- Example of Manufacturing Company
 - Sales
 - Shipment
 - Inventory, etc.

Integrated Data

- Data comes from several applications
- Problems of integration comes into play
 - File layout, encoding, field names, systems, schema, data heterogeneity are the issues
 - Bank example, variance: naming convention, attributes for data item, account no, account type, size, currency
- In addition to internal, external data sources
 - External companies data sharing
 - Websites
 - Others
- Removal of inconsistency
- So process of extraction, transformation & loading

Time variant

- Operational data has current values
- Comparative analysis is one of the best techniques for business performance evaluation
- Time is critical factor for comparative analysis
- Every data structure in DW contains time element
- In order to promote certain products, analyst has to know about current and historical values
- The advantages are
 - Allows for analysis of the past
 - Relates information to the present
 - Enables forecasts for the future

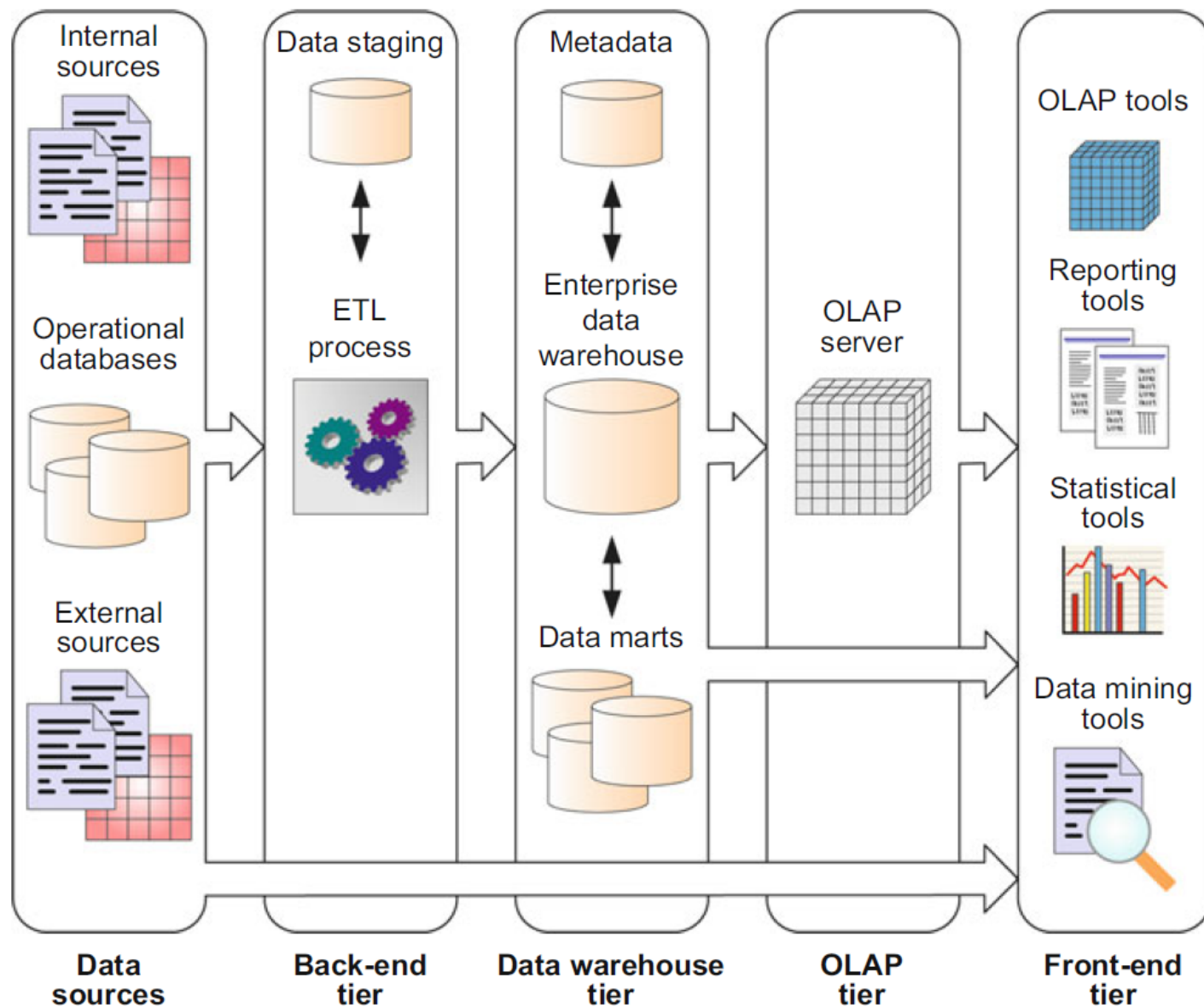
Non-volatile

- Data from operational systems are moved into DW after specific intervals
- Data is persistent/ not removed i.e. non volatile
- Every business transaction don't update in DW
- Data from DW is not deleted
- Data is neither changed by individual transactions
- Properties summary

Subject Oriented	Time-Variant	Non-Volatile
Organized along the lines of the subjects of the corporation. Typical subjects are customer, product, vendor and transaction.	Every record in the data warehouse has some form of time variancy attached to it.	Refers to the inability of data to be updated. Every record in the data warehouse is time stamped in one form or another.

Multidimensional Model

Typical DW architecture



Dimensional Model

- Data warehouses and OLAP systems are based on **multidimensional model**
- Dimensional modeling focuses subject-orientation, critical factors of business
- Critical factors are stored in **facts**

Dimensional Model

- Logical design technique for high performance
- Each model represent a subject in DW
- Is the modeling technique for storage

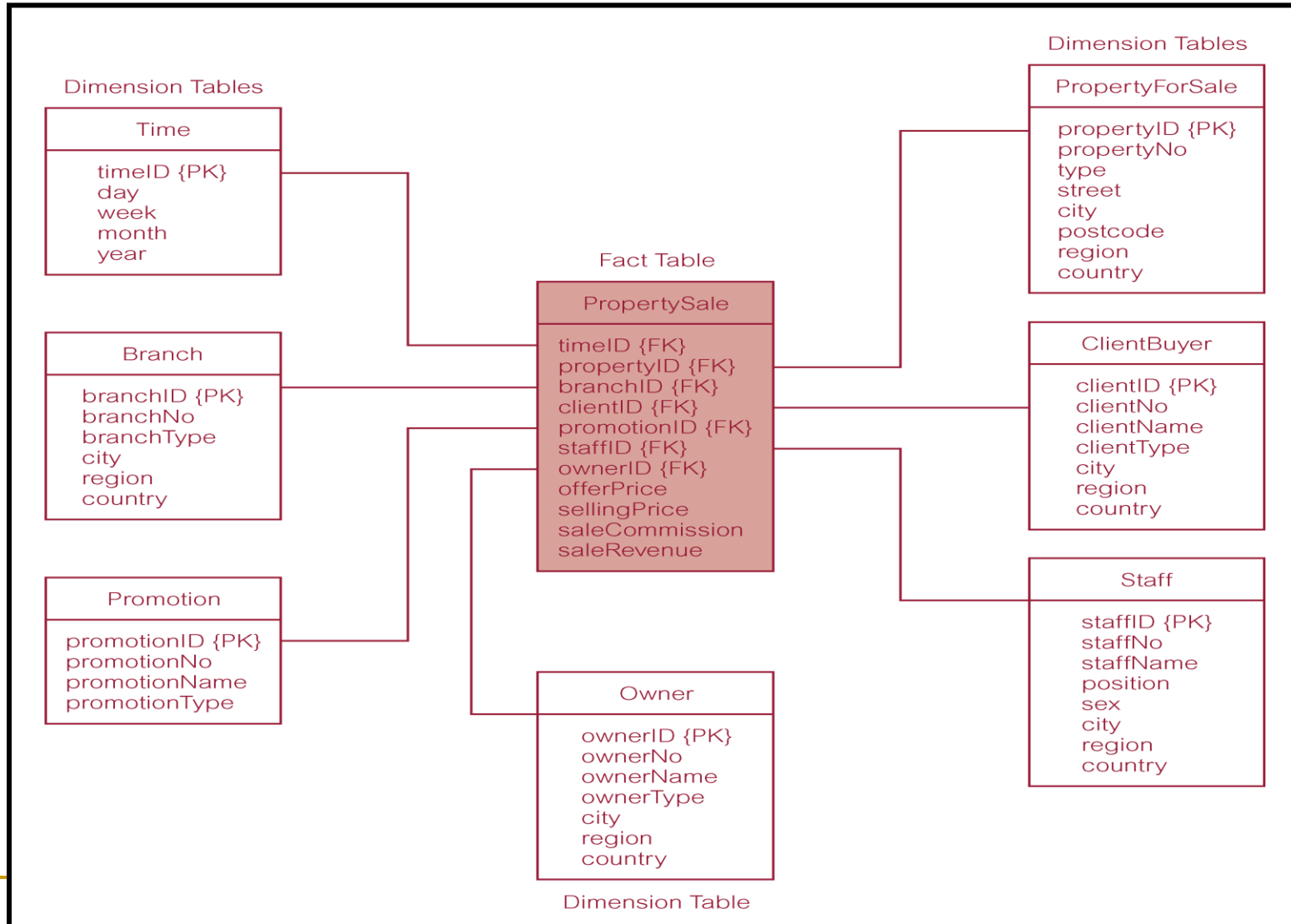
Dimensional Model

- Two important concepts
 - Fact
 - Numeric measurements, represent business activity/event
 - Are pre-computed, redundant
 - Example: Profit, quantity sold
 - Dimension
 - Qualifying characteristics, perspective to a fact
 - Example: date (Date, month, quarter, year), product(type, category)

Dimensional Model

- Every dimensional model (DM) is composed of one (or more) fact tables, and a set of smaller dimension tables.
- Look on **Fact table** through one (or more) **dimensions**.
 - What is the **sale amount** in **Consumer Product category**, for **elderly customers** in the **second quarter** of **2004**?
- Forms 'star-like' structure, which is called a star schema or star join.

A Typical Dimensional Model

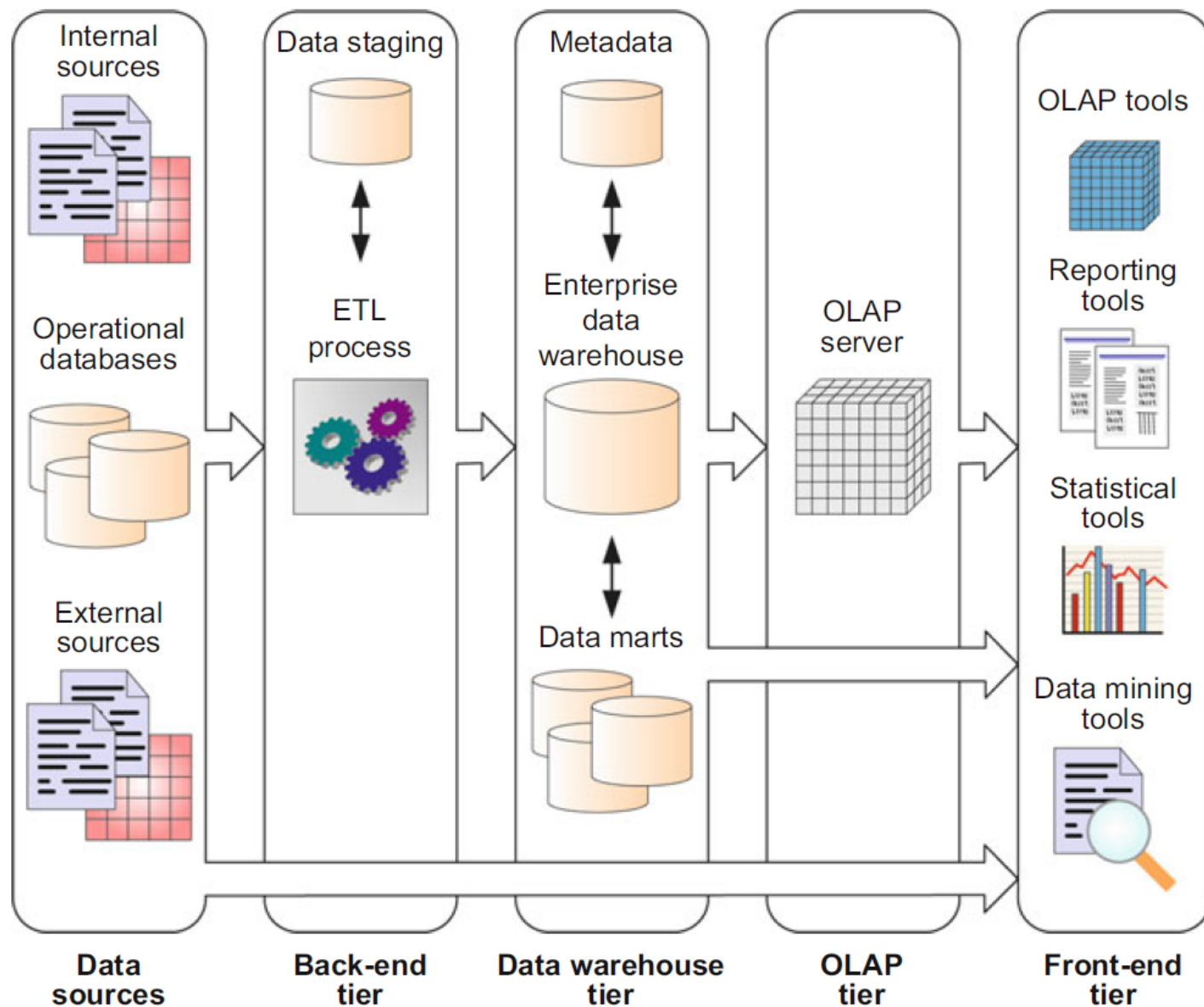


Operational DB vs DW

	Aspect	Operational databases	Data warehouses
1	User type	Operators, office employees	Managers, executives
2	Usage	Predictable, repetitive	Ad hoc, nonstructured
3	Data content	Current, detailed data	Historical, summarized data
4	Data organization	According to operational needs	According to analysis needs
5	Data structures	Optimized for small transactions	Optimized for complex queries
6	Access frequency	High	From medium to low
7	Access type	Read, insert, update, delete	Read, append only
8	Number of records per access	Few	Many
9	Response time	Short	Can be long
10	Concurrency level	High	Low
11	Lock utilization	Needed	Not needed
12	Update frequency	High	None
13	Data redundancy	Low (normalized tables)	High (denormalized tables)
14	Data modeling	UML, ER model	Multidimensional model

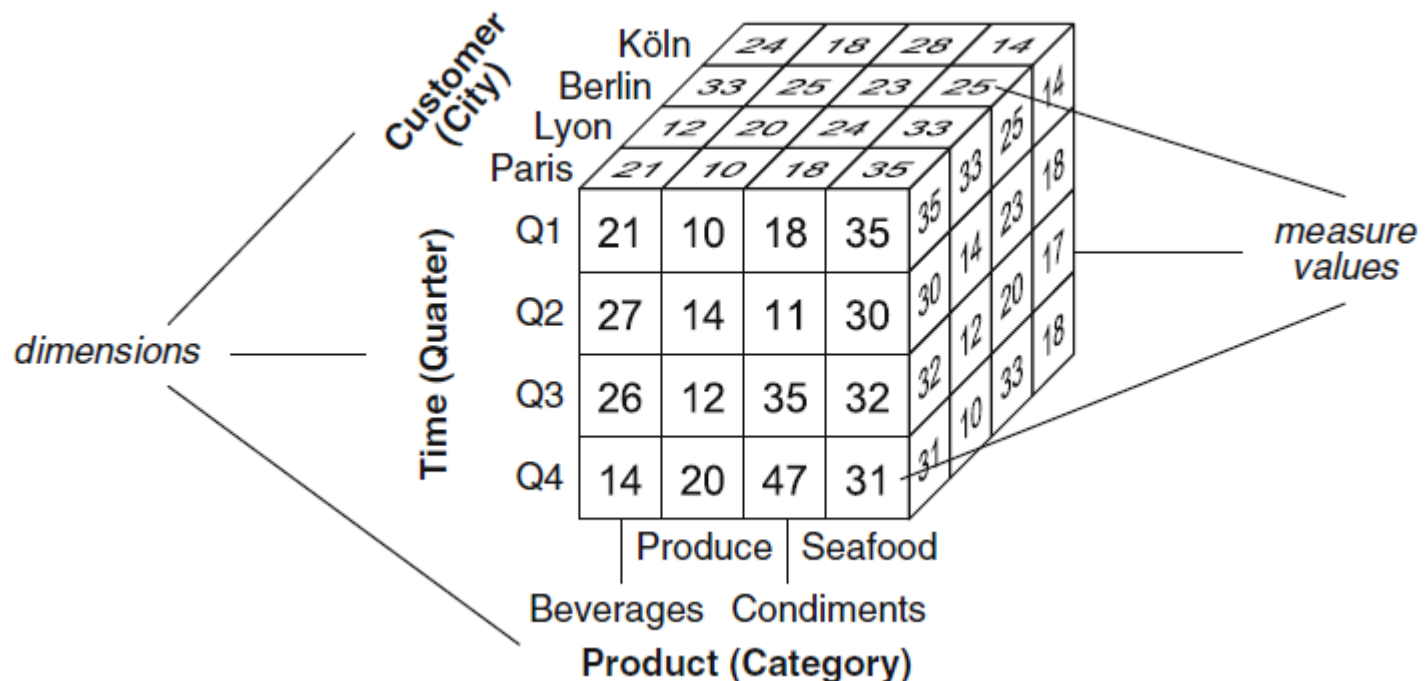
OLAP

Typical DW architecture



Data Cube

- A data cube is defined by **dimensions** and **facts**



Cube

- The cube has three dimensions
 - Product
 - Time
 - Customer city
- The cells of a data cube, or **facts**, have associated **numeric values**

OLAP Operations

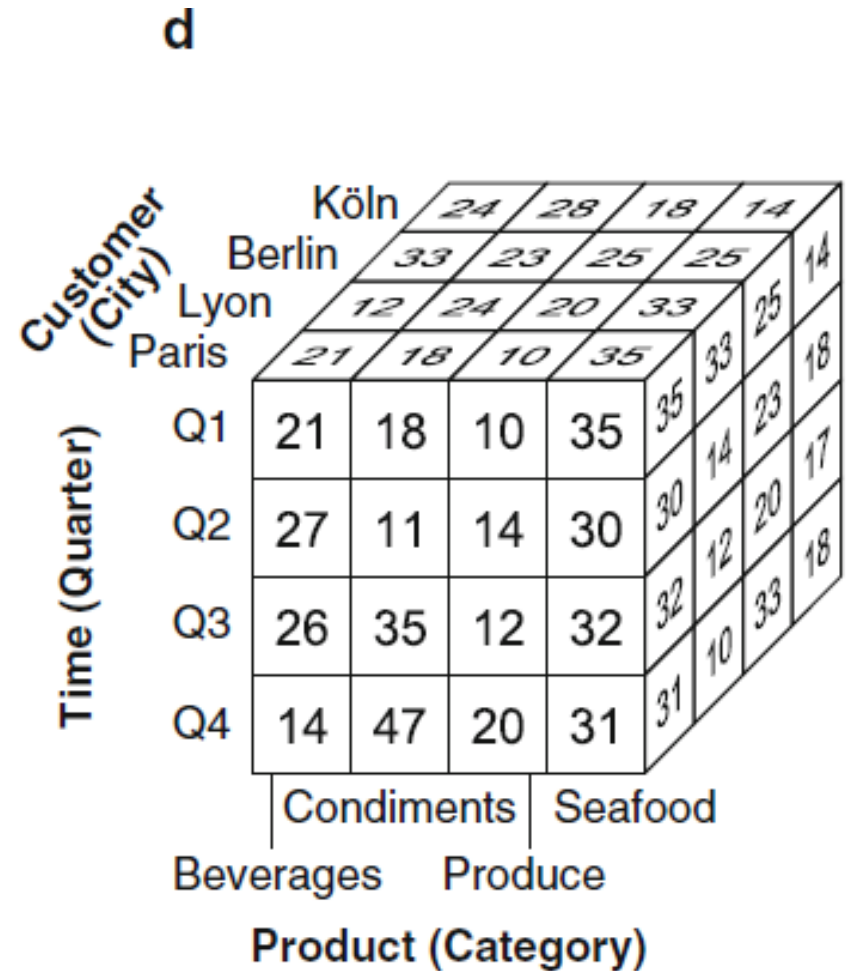
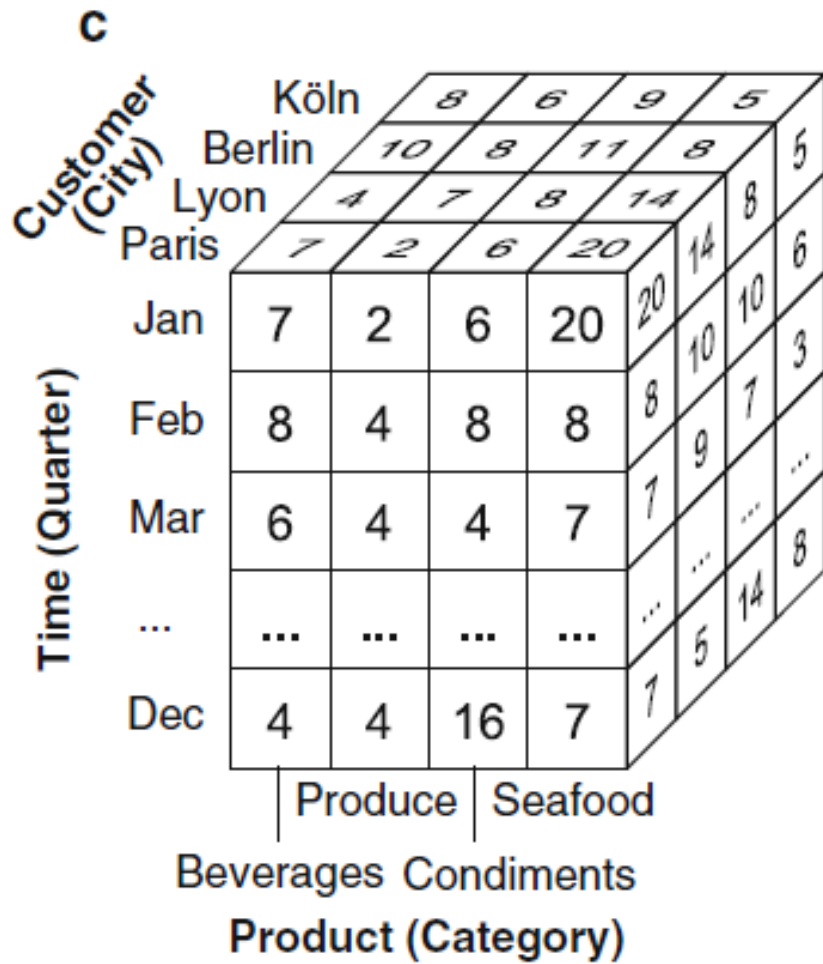
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Time (Quarter)	Customer (City)				Product (Category)			
	Köln							
	Berlin							
	Lyon							
	Paris							
Q1	21	10	18	35	35	33	25	14
Q2	27	14	11	30	30	14	23	17
Q3	26	12	35	32	32	12	20	18
Q4	14	20	47	31	31	10	33	18
					Produce	Seafood	Beverages	Condiments

b

Time (Quarter)	Customer (Country)				Product (Category)			
	Germany							
	France							
Q1	33	30	42	68	68	39	41	37
Q2	39	26	41	44	44	39	41	37
Q3	30	22	46	44	44	39	41	37
Q4	25	29	49	41	41	39	41	37
					Produce	Seafood	Beverages	Condiments

OLAP Operations



OLAP Operations

e

Product (Category)	Seafood	35	30	32	31	20	47	31	
	Condiments	18	11	35	47				
	Produce	10	14	12	20				
	Beverages	21	27	26	14				
		14	17	21	10				
Customer (City)	Paris	21	27	26	14	13	17	21	10
	Lyon	12	14	11	13	32	28	20	33
	Berlin	33	28	35	32	19	47	18	
	Köln	24	23	25	18				
		Q1	Q2	Q3	Q4				
		Time (Quarter)							

Business Intelligence Tools

BI tools

- Microsoft
- Oracle
- IBM
- Teradata
- SAP
- Microstrategy
- Targit

Microsoft SQL Server tools

- Database Engine
- Integration services (SSIS)
- Analysis Services & Reporting Services